University of Sheffield

Employment Dynamics in Yorkshire and the Humber: Trends, COVID-19 Impacts and Future Projections

UCHE DUMZO-AJUFO

1 INTRODUCTION AND AIMS

1.1 CONTEXT AND MOTIVATION

Employment patterns offer insight into regional economic conditions, revealing how its structure is evolving, how resilient its labour market is and how it reacts to external pressures such as financial downturns or global pandemics (Harari & Ward, 2022). In the UK, sectoral employment has shifted over the past decade due to technological advancements, globalisation, demographic changes and evolving policy priorities (ONS, 2023). Analysing these patterns at a regional level is essential for informing local policy as well as investment and workforce development strategies.

Yorkshire and the Humber (hereafter "Yorkshire") has one of the UK's most diverse economies, combining traditional sectors, like manufacturing and construction, with growing service and digital industries. Recent years have seen significant transitions, including the decline of heavy industry, growth in healthcare and education and disruption caused by COVID-19 (Kaye, 2020). Examining industry employment trends helps assess the region's adaptability, resilience and ability to capitalise on emerging economic opportunities (Barzotto et al., 2025).

This study uses the Business Register and Employment Survey (BRES), which provides annual employment estimates across UK industries. Focusing on Yorkshire from 2015 to 2024, it identifies long-term trends, evaluates the impact of COVID-19 and outlines potential changes through to 2030. The findings support evidence-based decisions for policymakers, businesses and workforce planners by highlighting region-specific employment dynamics.

1.2 LITERATURE REVIEW

Research on UK regional labour markets highlights differences in industrial structure and employment growth. Evidence from the Office for National Statistics (ONS, 2023) and regional economic observatories show that northern regions, including Yorkshire, still depend on manufacturing and public services, while London and the South East dominate in knowledge-intensive and digital sectors.

Work on regional resilience (Martin & Sunley, 2015) argues that adaptability depends on the diversity and flexibility of local industry bases. Regions that combine traditional and emerging sectors tend to recover more effectively from economic recessions. In Yorkshire, the expansion of health and education sectors has been linked to demographic change and policy investment (Beatty & Fothergill, 2021), while growth in digital and professional services reflects the gradual shift towards a knowledge-based economy (Aparicio et al., 2021).

The COVID-19 pandemic widened existing regional differences. ONS research (2020) reported sharp short term job losses in hospitality, retail and construction, while healthcare, information technology and logistics recorded growth. Forecasting studies, such as PwC (2023), highlight the need for detailed regional analysis to understand post-pandemic recovery paths and future employment patterns.

Despite extensive literature, there is limited region-specific work that brings together recent longitudinal employment data with predictive modelling to track changes in Yorkshire's industrial composition. Much existing research focuses on national trends or short-term pandemic impacts, offering limited insight into how sectors may evolve locally. There is also a lack of studies that consider both traditional and emerging sectors to evaluate regional economic, and few analyses cover multi-year periods suitable for reliable projection.

This study addresses these gaps by applying statistical analysis and forecasting to BRES data for Yorkshire. It delivers an up-to-date assessment of industry performance and generates projections that illustrate long-term trends and potential future developments in the region's employment landscape.

1.3 AIMS AND RESEARCH QUESTIONS

The aim of this research is to analyse employment trends across Yorkshire from 2015 to 2024, assess the impact of the COVID-19 pandemic and forecast sectoral employment changes up to 2030 using regression-based projections. The research questions are:

1. **RQ1:** Which industries have shown the most consistent employment growth between 2015 and 2024?
2. **RQ2:** How did the COVID-19 pandemic (2019-2021) affect employment levels across different industries in Yorkshire?
3. **RQ3:** Based on historical trends, which industries are projected to grow or decline by 2030?

These questions guide the assessment of long-term trends, pandemic impacts and future workforce dynamics.

2 METHODOLOGY

2.1 DATA SOURCE AND COLLECTION

This study uses data from the BRES, which provides detailed annual employment estimates for UK industries. The dataset covers Yorkshire from 2015 to 2024 and an Industry column and separate yearly columns. Each year contains employee counts rounded to the nearest thousand and accompanying flags that record. This source was chosen for its reliable, longitudinal record of regional employment, enabling the analysis of long-term trends, the effects of major events like COVID-19 and anticipated future employment patterns.

2.2 DATA PRE-PROCESSING AND CLEANING

The raw data required restructuring before analysis. Flag columns were removed as they did not contribute substantive information. Industry names were cleaned using regular expressions to remove numerical prefixes and brackets. The dataset was reshaped from wide to long format, producing three variables: Industry, Year and Employees. Year was converted to numeric and employee counts were checked for missing or invalid entries. These steps produced a tidy dataset suitable for descriptive and predictive analysis. All processing was completed in R using readxl, tidyverse, stringr, ggplot2, ggrepel and scales.

2.3 DATA ANALYSIS TECHNIQUES

Analysis followed the three research questions. Descriptive statistics and line plots summarised overall and sector specific employment trends from 2015 to 2024. Percentage change from 2019 to 2021 captured the effects of COVID-19. For forecasting, linear regression models were fitted for each industry with Year as the predictor and Employees as the outcome. These models generated estimates for 2030 and associated growth rates. Visualisations created with ggplot2 supported interpretation of historical trends and projections.

2.4 Data Quality Considerations

BRES is considered robust for regional analysis, although sampling variation may affect smaller industries. The selected variables contained no significant missing data, and reporting was consistent across the 2015 to 2024 period. Regression based forecasts assume that past trends continue, so they may not reflect future shocks, policy shifts or structural changes (Casson and Farmer, 2014). These constraints were taken into account when interpreting results.

2.5 Visualisation

Visualisations played a central role in presenting findings. Line plots showed employment trends for all industries and for the five largest sectors. Bar charts illustrated overall growth, pandemic related changes and projected growth to 2030. Clear labelling and consistent formatting improved readability. All figures were exported in high resolution for inclusion in the report.

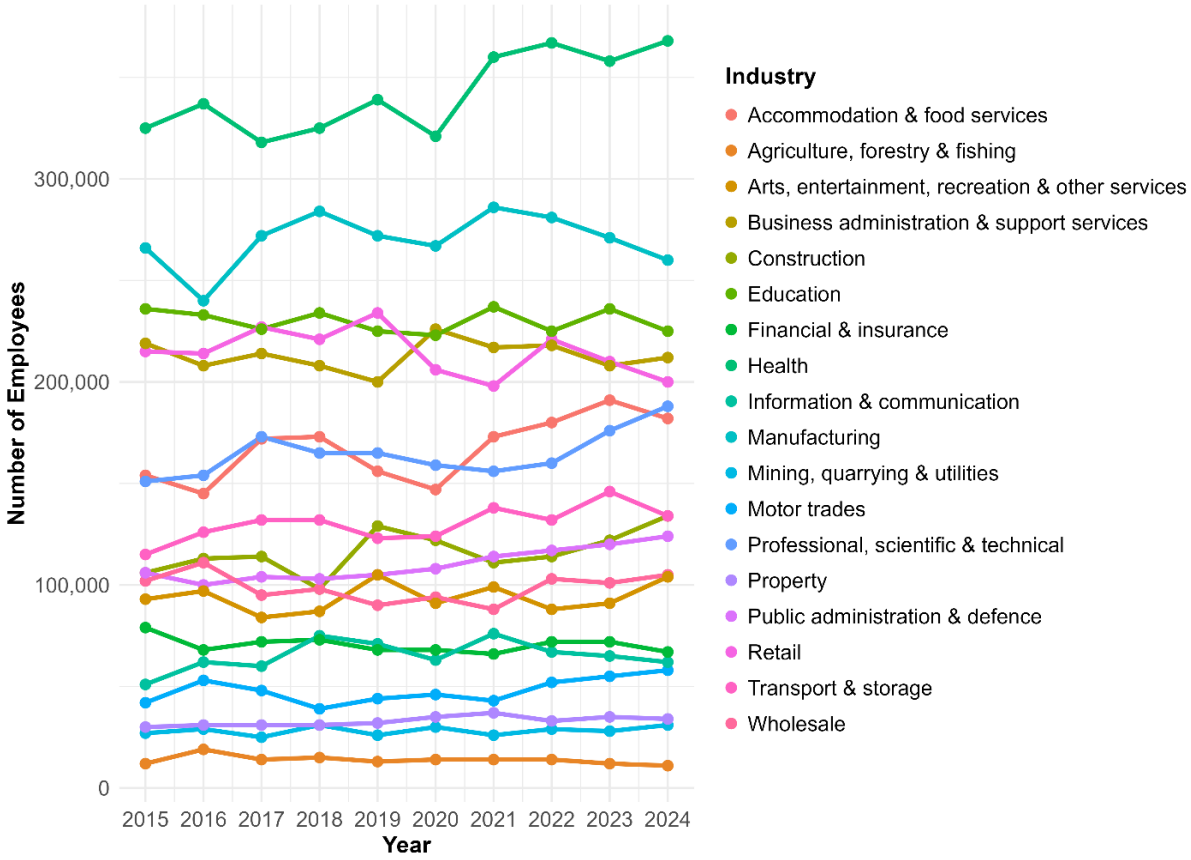
3 RESULTS

3.1 OVERVIEW OF EMPLOYMENT TRENDS

Employment across Yorkshire between 2015 and 2024 displayed stability in traditional sectors and growth in emerging ones. Health and Manufacturing consistently employed the largest portions of the workforce. Health rose steadily from about 325,000 employees in 2015 to 367,000 in 2022. Manufacturing fluctuated between 240,000 and 286,000. Other large sectors, including Education and Retail, showed comparatively smaller changes, reflecting overall stability. Information and Communication recorded gradual growth. Figure 1 shows trends across all industries and Figure 2 highlights the top five sectors.

Figure 1

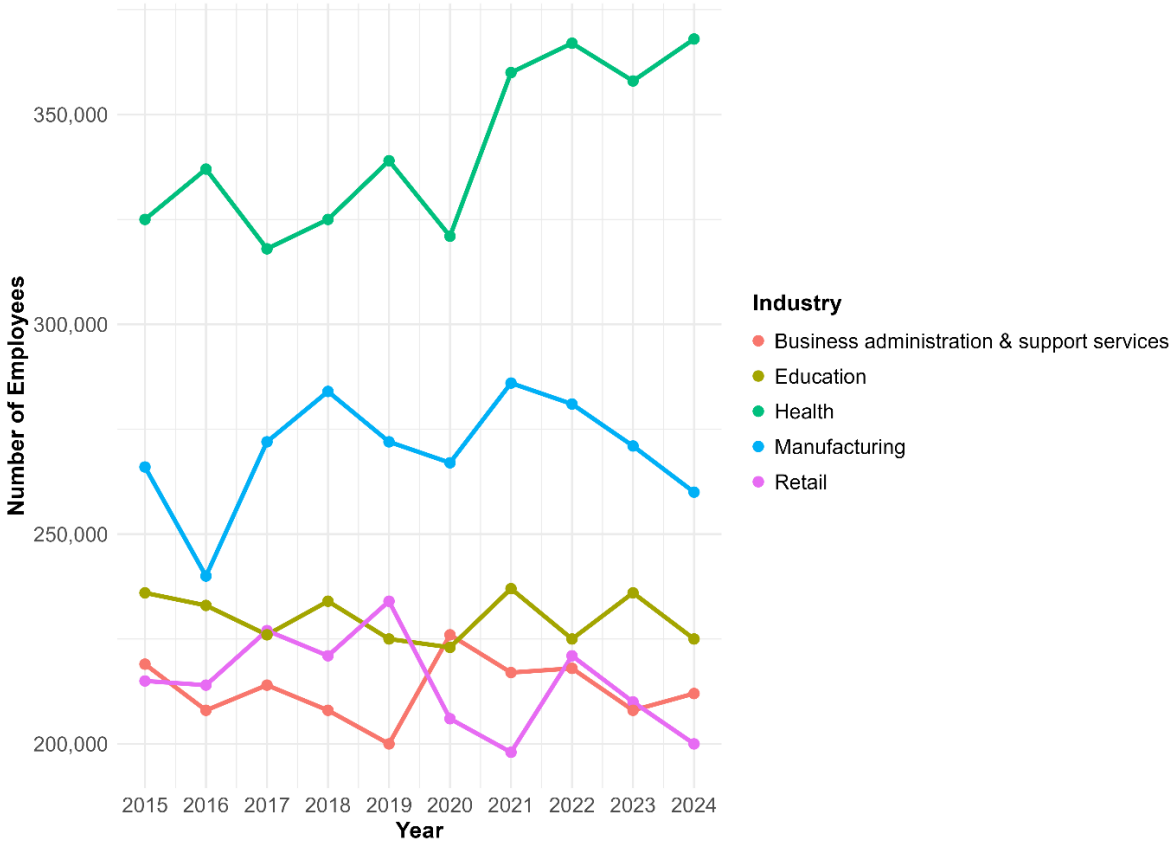
Employment trends across all industries in Yorkshire and the Humber (2015-2024)



Note. The line plot illustrates the total number of employees (y-axis) across years (x-axis) for each industry included in the dataset. Each coloured line represents a distinct industry, showing broad differences in employment scale and variation over time.

Figure 2

Employment trends of the top five industries in Yorkshire and the Humber (2015-2024)



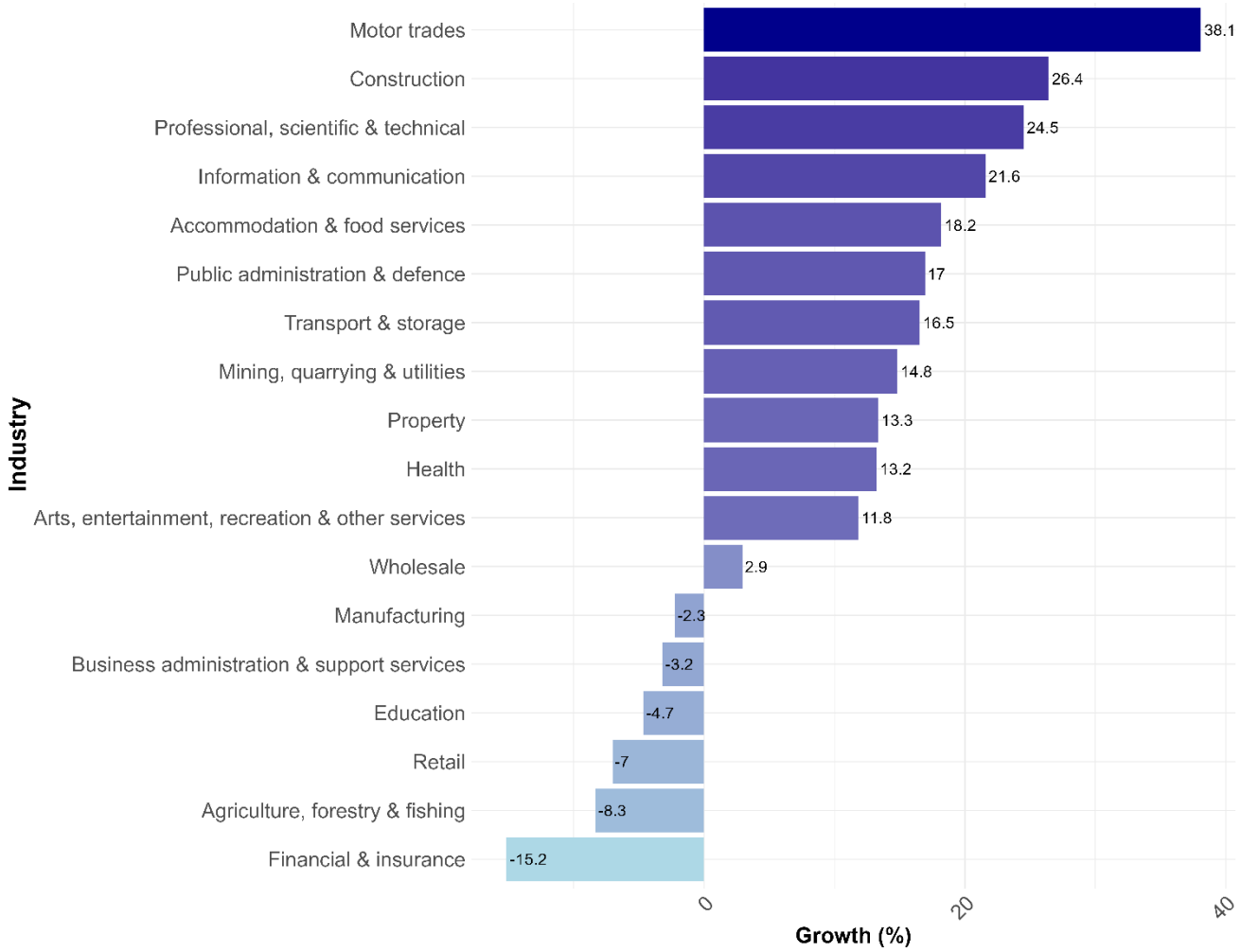
Note. The line plot shows the number of employees (y-axis) across years (x-axis). Health is shown in green, Manufacturing in blue, Business Administration & Support Services in red, Education in yellow and Retail in pink.

3.2 Industries with Consistent Growth (RQ1)

Employment growth between 2015 and 2024 varied widely. Motor Trades experienced the fastest proportional growth at 38.1%, followed by Construction (26.4%), Professional, Scientific & Technical services (24.5%) and Information & Communication (21.6%). Moderate growth was observed in Accommodation & Food Services (18.2%) and Health (13.2%). In contrast, several traditionally large industries experienced declines, including Manufacturing (-2.3%), Business Administration & Support Services (-3.2%), Retail, Education, Agriculture and Financial & Insurance, which declined by up to -15.2%. These patterns are summarised in Figure 3, which presents the percentage employment growth by industry across the period.

Figure 3

Percentage employment growth by industry in Yorkshire and the Humber (2015-2024)



Note. Industries are ordered from lowest to highest growth, with bars coloured along a gradient from light blue (lower growth) to dark blue (higher growth). Values are labelled to indicate exact percentage change over the period.

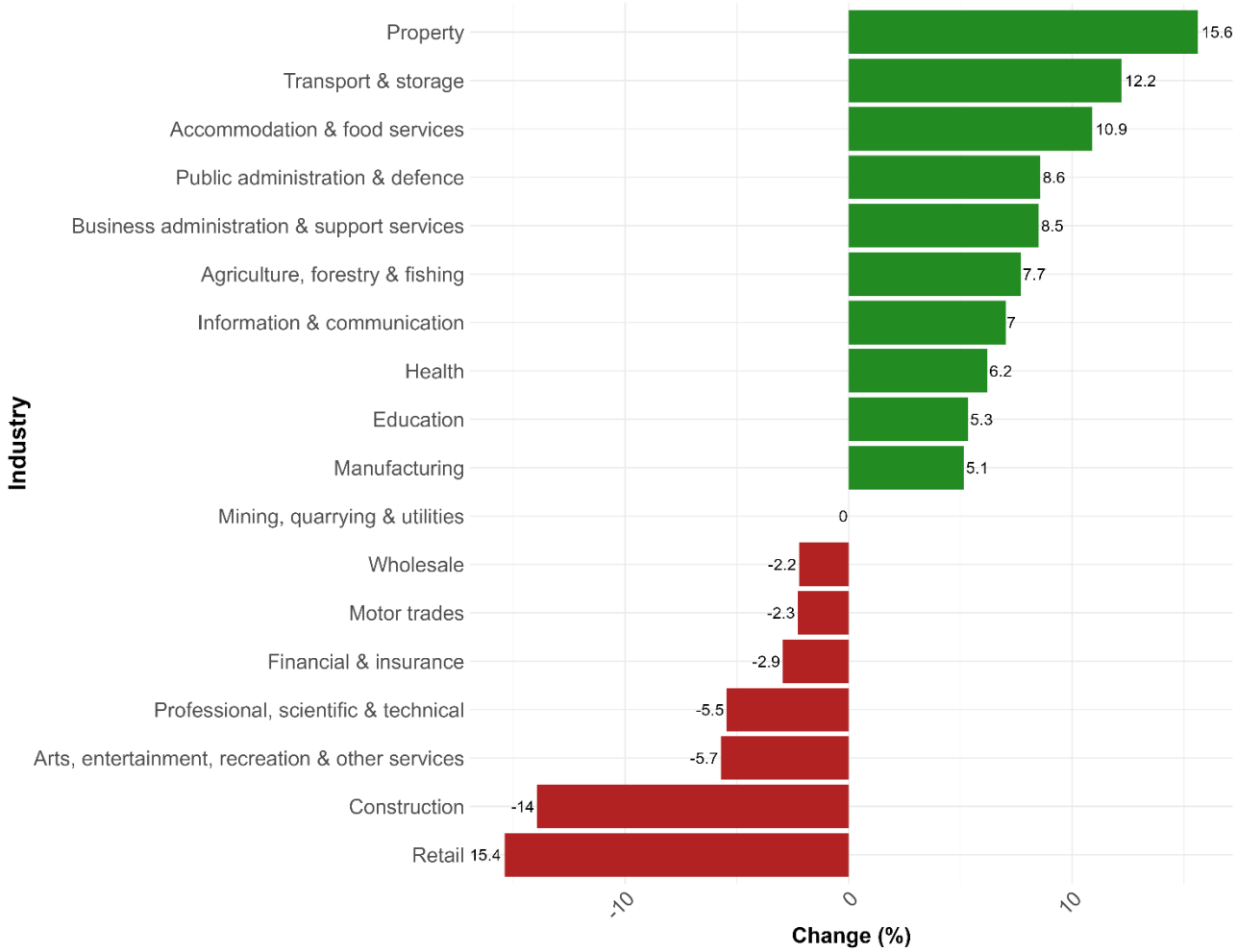
3.3 COVID-19 Employment Impact (RQ2)

Between 2019 and 2021, the pandemic had uneven effects across sectors. The Property sector recorded the largest increase at +15.6%, followed by Transport & Storage (+12.2%) and Accommodation & Food Services (+10.9%). Public Administration & Defence (+8.6%) and Business Administration & Support Services (+8.5%) also expanded, while Health (+6.2%), Education (+5.3%), Manufacturing (+5.1%) and several other sectors showed moderate gains. Mining, Quarrying & Utilities remained stable with no significant change.

On the contrary, Retail (-15.4%) and Construction (-14.0%) experienced the largest declines, with smaller contractions in Professional, Scientific & Technical (-5.5%), Arts, Entertainment, Recreation & Other Services (-5.7%) and Financial & Insurance (-2.9%). The mixed pattern of growth and decline across sectors is illustrated in Figure 4, highlighting the differing resilience of industries during the pandemic.

Figure 4

Employment change in Yorkshire industries during the COVID-19 pandemic (2019-2021)



Note. Positive changes are shown in green and negative changes in red, with value labels indicating the percentage change for each industry. Industries are ordered from largest to smallest change.

3.4 PREDICTING FUTURE EMPLOYMENT TO 2030 (RQ3)

Table 1 summarises predicted employment figures by industry from the linear regression model, showing 2024 baseline values, 2030 forecasts and projected percentage growth. The largest increases are expected in Information & Communication (+20.6%), Property (+14.5%), Accommodation & Food Services (+13.1%) and Transport & Storage (+12.8%). Moderate growth is projected for Public Administration & Defence (+9.5%), Manufacturing (+8.9%) and Health (+7.9%), while Motor Trades (+3.8%) and Construction (+3.3%) show smaller increases. Education (+0.1%) and Business Administration & Support (+0.8%) are predicted to remain largely stable.

Sectors expected to decline include Mining, Quarrying & Utilities (-1.2%), Retail (-1.9%), Arts, Entertainment & Recreation (-3.7%), Financial & Insurance (-4.4%), Wholesale (-7.8%) and Agriculture, Forestry & Fishing (-9.3%). These forecasts are illustrated in Figure 5, which shows predicted employment growth percentages; Figure 6, which combines historical trends with 2030 predictions; and Figure 7, which displays the projected distribution of employment across industries in 2030.

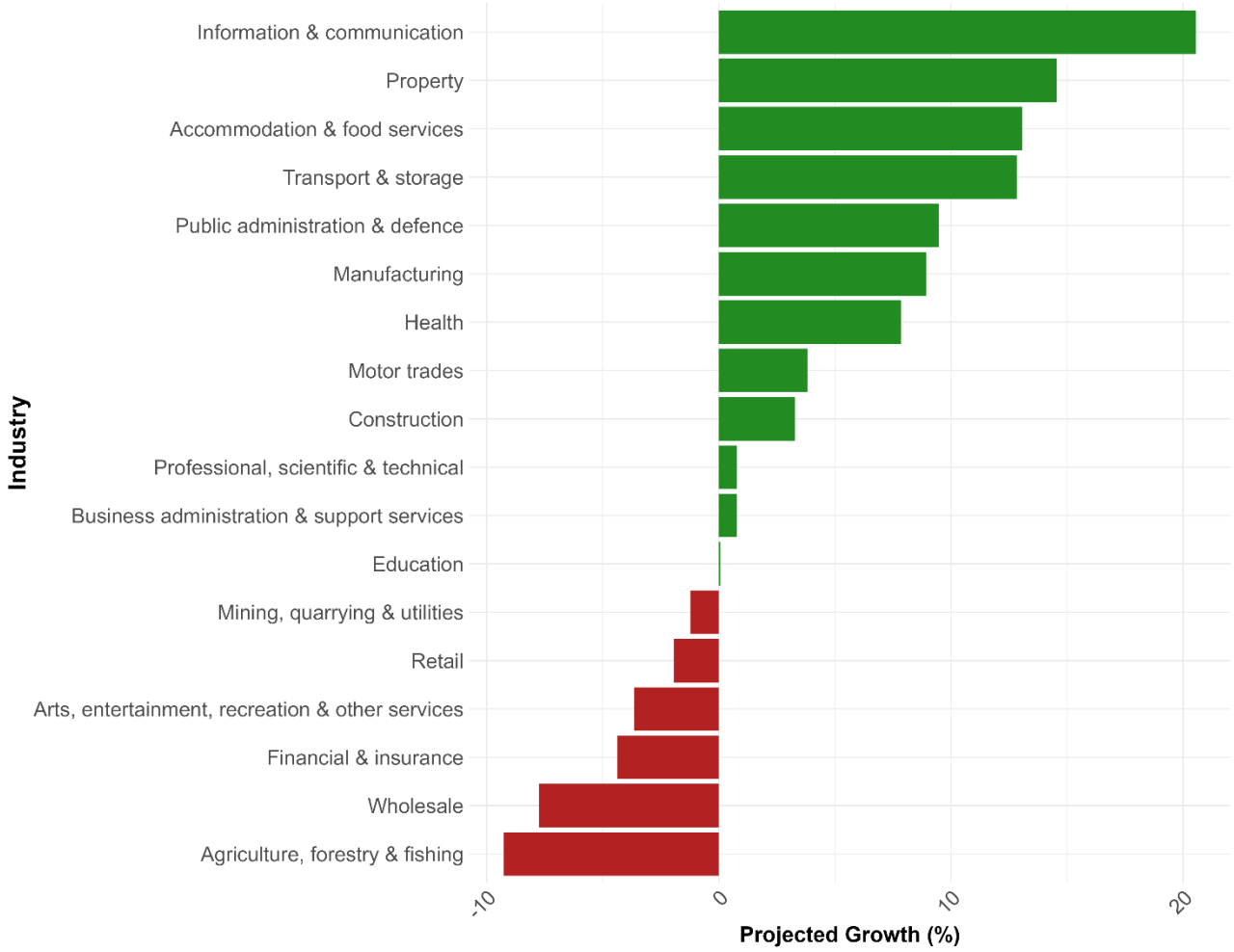
Table 1

Projected employment by industry in Yorkshire and the Humber (2024-2030)

Rank	Industry	Employees (2024)	Predicted (2030)	Projected Growth (%)
1	Information & Communication	62,000	74,745	+20.6
2	Property	34,000	38,945	+14.5
3	Accommodation & Food Services	182,000	205,800	+13.1
4	Transport & Storage	134,000	151,200	+12.8
5	Public Administration & Defence	124,000	135,745	+9.5
6	Manufacturing	260,000	283,200	+8.9
7	Health	368,000	396,909	+7.9
8	Motor Trades	58,000	60,218	+3.8
9	Construction	134,000	138,382	+3.3
10	Professional, Scientific & Technical	188,000	189,455	+0.8
11	Business Administration & Support	212,000	213,636	+0.8
12	Education	225,000	225,164	+0.1
13	Mining, Quarrying & Utilities	31,000	30,618	-1.2
14	Retail	200,000	196,145	-1.9
15	Arts, Entertainment & Recreation	104,000	100,200	-3.7
16	Financial & Insurance	67,000	64,073	-4.4
17	Wholesale	105,000	96,855	-7.8
18	Agriculture, Forestry & Fishing	11,000	9,982	-9.3

Figure 5

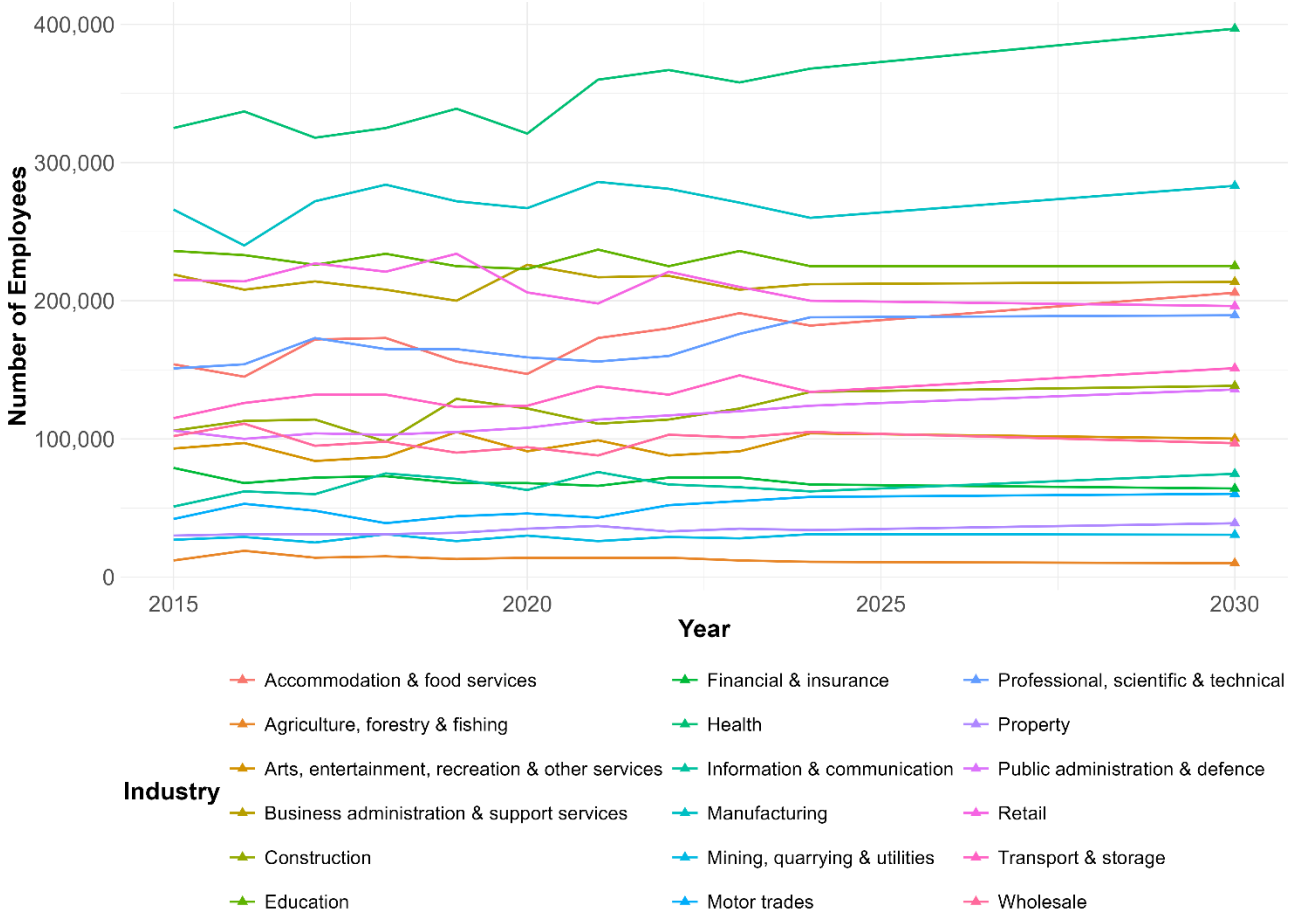
Predicted employment growth by Yorkshire industry between 2024 and 2030



Note. Positive projected growth is shown in green, while negative growth is in red. Industries are ordered by projected growth percentage, with value labels indicating the percentage change.

Figure 6

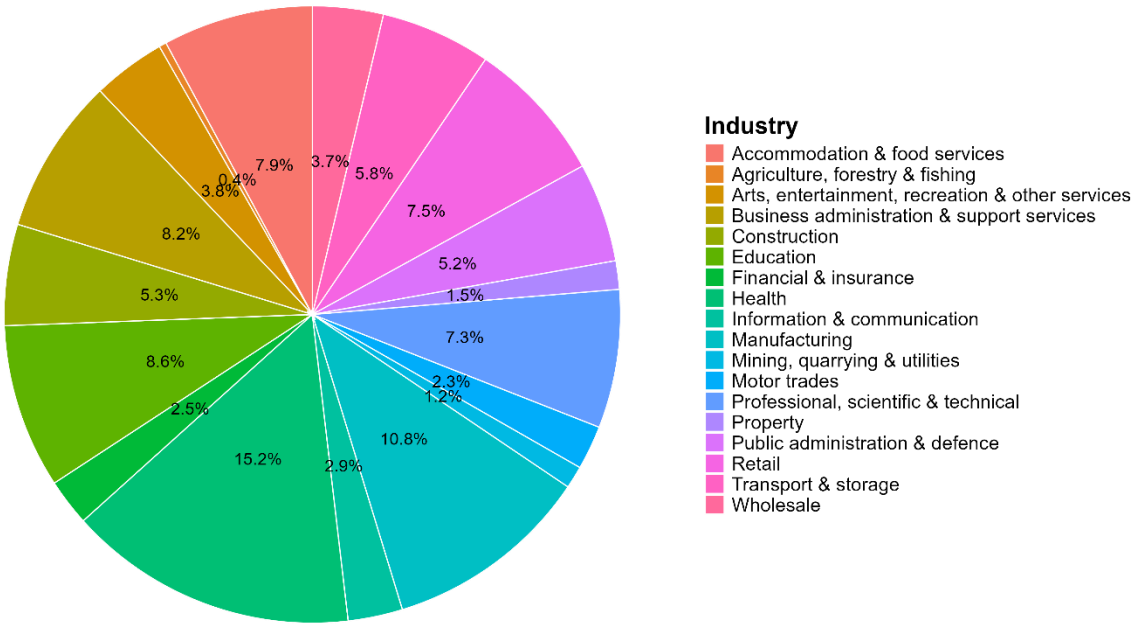
Historical employment trends (2015-2024) with predicted employment for 2030 across Yorkshire industries



Note. The line plot shows historical data, while the 2030 predictions are highlighted with triangle markers to indicate expected future employment levels.

Figure 7

Predicted distribution of employment across industries in Yorkshire for 2030



Note. The pie chart shows each industry's share of total employment, highlighting the relative sizes of sectors based on projected growth trends.

4 DISCUSSION

4.1 INTERPRETING EMPLOYMENT TRENDS

The analysis of employment in Yorkshire from 2015 to 2024 demonstrates stability in traditional sectors and gradual diversification towards emerging industries. Health and Manufacturing continue to dominate in absolute employment, reflecting a sustained demand for healthcare services and ongoing industrial activity. Sectors such as Information & Communication, Professional, Scientific & Technical services and Accommodation & Food Services show higher proportional growth, signalling a shift towards knowledge-based and service-driven employment. Overall, these findings show that while established industries remain critical to regional employment, growth is increasingly centred on technology and services.

4.2 INDUSTRIES WITH CONSISTENT GROWTH (RQ1)

Industries showing consistent growth over the 2015-2024 period, particularly Motor Trades, Construction, Professional & Technical Services and Information & Communication, represent developing strengths within the regional economy. Figure 3 demonstrates that these sectors experienced the largest proportional increases, suggesting that they are key drivers of diversification. Conversely, traditionally large industries such as Retail, Education and Business Administration & Support Services displayed slower growth or modest decline, indicating stabilisation rather than significant contraction. This pattern emphasises the ongoing shift in the regional employment landscape, with high-growth sectors complementing the continued presence of established industries.

4.3 COVID-19 Employment Impact (RQ2)

The COVID-19 pandemic had an uneven impact on employment across industries, as shown in Figure 4. Sectors such as Property, Transport & Storage and Accommodation & Food Services demonstrated short-term growth, reflecting adaptability to changes in demand and operational flexibility. In contrast, Construction, Retail and Arts-related industries experienced notable declines due to restrictions on physical operations and reduced consumer activity. These findings highlight the vulnerability of industries reliant on face-to-face interactions and discretionary spending, while digitally enabled and essential services were able to maintain or even expand employment. The pandemic therefore heightened structural differences in sectoral resilience within the region.

4.4 Predicting Future Employment (RQ3)

Projections to 2030, based on the linear regression models of historical data, indicate continuity and change in the regional workforce. Health and Manufacturing are expected to maintain dominance in absolute employment, while sectors such as Information & Communication and Property are forecast to experience substantial proportional growth, shown in Figures 5 and 6. Some traditionally large sectors, including Wholesale, Agriculture and Financial & Insurance, are predicted to decline modestly. Figure 7 illustrates the projected distribution of employment in 2030, showing the increasing influence of high-growth sectors. These trends suggest that regional policymakers and workforce planners should focus on aligning skills development with growing industries while supporting sectors facing stagnation or decline.

4.5 Policy and Strategic Considerations

The findings carry several implications for regional economic planning. Targeted skills development in digital, technical and professional domains can help ensure the workforce meets the demand of expanding sectors. Sustaining capacity in established industries such as Health and Manufacturing is vital for economic stability, while support for declining sectors can mitigate the risks associated with structural changes. Promoting diversification across industries reduces overreliance on a single sector and boosts the resilience and adaptability of the regional economy (Yaman & Tuncel, 2025). The combination of historical trends and 2030 projections gives evidence to guide investment, training and policy priorities within Yorkshire.

4.6 Limitations and Future Research

This study is constrained by several limitations. Employment data were analysed at the industry level, which may obscure variations at the occupational level. Linear regression projections assume the continuation of historical trends and do not account for potential structural changes, policy interventions or economic shocks (Xiao & Jin, 2021). Additionally, the regional focus does not consider broader national or global influences, such as migration, trade or technological disruption (Baldwin et al., 2025). Future research could incorporate occupational-level data, scenario-based predictive modelling and analyses linking regional employment dynamics to wider economic and demographic factors. These approaches would provide a more nuanced understanding of Yorkshire's labour market and strengthen the evidence base for strategic planning.

5 R CODE AND GITHUB PAGES

All the materials for this project are available on my GitHub so that the analysis is fully transparent and reproducible. The repository (yorkshire-employment-analysis) includes the workflow used in the project, along with the data and generated outputs.

GitHub profile

Project repository

The repository is structured as follows:

- **intro_ds_code.R**
    - This is the main script containing all data cleaning steps, transformations, visualisations and the forecasting methods.
- **bres_yorkshire_industry.xlsx**
    - The dataset used in the analysis. This file is loaded directly by the script, making the code self-contained and simple to run.
- **Plots folder**
    - This folder contains all charts generated during the analysis, including trend lines, growth comparisons, COVID-19 impact visualisations and 2030 predictions. These images are used throughout the report to support the findings.
- **README.md**
    - This file provides a short overview of the project, including the aim of the investigation, a summary of the methods used and a description of the repository structure. It also contains instructions for running the analysis.

To run the code, users only need to install the required R packages listed in the README, then run intro_ds_code.R in sequence. The script automatically loads the dataset, performs all preprocessing steps and saves the output figures to the Plots folder.

The full version of the R script included on GitHub has also been provided in Appendix A, along with screenshots of the GitHub profile and repository as required.

6 CONCLUSIONS

6.1 REFLECTIONS

6.1.1 Summary of Key Findings

This study examined employment trends across Yorkshire from 2015 to 2024, assessed the impact of COVID-19 and produced forecasts to 2030. Traditional sectors such as Health and Manufacturing remain central to the region's employment, showing long-term stability and strong absolute workforce sizes. Emerging sectors, including Information & Communication, Professional & Technical Services and Accommodation & Food Services, are driving proportional growth, highlighting a gradual shift towards knowledge and service-based employment. The COVID-19 pandemic affected industries unevenly, with digitally enabled and essential services proving more resilient, while sectors reliant on physical presence experienced notable declines. Projections to 2030 indicate continued growth in high-demand sectors, alongside modest declines in some traditional industries, suggesting the need for strategic workforce planning.

6.1.2 Broader Implications

These findings underline the need for targeted skills development, focused policy support and selective investment. By aligning workforce capabilities with emerging sectoral demand, regional stakeholders can foster economic resilience and diversification. Maintaining capacity in established industries allows for stability, while supporting sectors facing stagnation or decline helps mitigate the risks of structural shifts. This analysis draws attention to the potential for a balanced regional economy in which traditional sectors coexist with high-growth, technology and service-oriented industries.

6.1.3 Reflection on Limitations and Future Work

While the study provides valuable insights, it is constrained by its reliance on industry-level data, assumptions inherent in linear forecasting and the exclusion of broader national or global economic influences. Future research could address these limitations by incorporating occupational-level data, scenario-based modelling that accounts for policy and technological changes and analyses linking regional trends with wider economic and demographic factors. Such approaches would strengthen understanding of Yorkshire's evolving labour market and strengthen the evidence base for strategic decision-making.

6.1.4 Lessons Learned

This project demonstrates the power of combining historical data analysis, predictive modelling and visualisation to understand regional employment dynamics. It shows the value of interpreting quantitative results within socioeconomic contexts and the importance of clear communication in conveying insights. Overall, the work reinforces the role of data-driven approaches in informing workforce planning, policy development and sustainable regional growth.

6.2 ENGAGEMENT

6.2.1 Insights from Industry and Alumni Events

Participation in industry and alumni events provided practical perspectives on real-world data science applications. Discussions with professionals, including alumni in higher education analytics, highlighted the growing demand for predictive modelling, economic analytics and evidence-based decision-making across sectors. These insights emphasised that employment and student data must be interpreted within local policy, demographic trends and sector-specific factors such as automation, digitalisation and ageing populations.

Alumni also stressed the importance of communicating findings to non-technical stakeholders. Just as dashboards translate complex enrolment and attainment data into actionable insights for university leaders, regional employment analysis must present trends clearly to policymakers, workforce planners and business leaders. Understanding stakeholder questions ensures that the analysis informs skills development, investment and resource allocation.

6.2.2 Broader Understanding of Data Science Practice

Engagement activities and project work broadened my view of data science as both technical and strategic. Beyond coding, modelling and data cleaning, effective data science requires problem framing, reproducibility, transparency and clear visualisation. Professional examples demonstrated that iterative workflows, like merging, cleaning and modelling data in response to feedback, are essential for robust outputs.

The experience highlighted that impactful data science combines technical proficiency with contextual knowledge and communication skills. Success depends on framing the right questions, designing interpretable models and translating findings into narratives that guide strategic decisions. These lessons show the value of integrating analytical precision with domain understanding to support policy, planning and sustainable outcomes.

7 REFERENCES

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8 APPENDIX

8.1 APPENDIX A - R CODE

```
#=========================================================#
#      Yorkshire Industry Employment Analysis (2015-2024)#
#=========================================================#


#-------------------------#
#       Libraries         #
#-------------------------#
library(readxl)
library(tidyverse)
library(stringr)
library(ggpubr)
library(ggrepel)
library(scales)


#-------------------------#
#       Loading Data      #
#-------------------------#
bres <- read_excel("bres_yorkshire_industry.xlsx") %>%
  select(-contains("Flags")) %>%  # Removes unnecessary columns
  mutate(Industry = str_replace(Industry, "^\\d+\\s*:\\s*", "") %>%
           str_replace("\\s*\\([^)]*\\)$", ""))  # Clean names


bres_long <- bres %>%
  pivot_longer(cols = `2015`:`2024`, names_to = "Year", values_to = "Employees") %>%
  mutate(Year = as.numeric(Year))
```

```
#-------------------------#
#   Exploratory Analysis  #
#-------------------------#
summary(bres_long$Employees)
length(unique(bres_long$Industry))


regional_trend <- bres_long %>%
  group_by(Year) %>%
  summarise(TotalEmployment = sum(Employees, na.rm = TRUE))


#-------------------------#
#   Employment Trends     #
#-------------------------#
employment_trends_all_industries <- ggplot(bres_long, aes(Year, Employees, colour =
Industry)) +
  geom_line(size = 2) +
  geom_point(size = 4) +
  scale_y_continuous(labels = comma) +
  scale_x_continuous(breaks = 2015:2024) +
  labs(title = "Employment Trends by Industry (2015-2024)",
       x = "Year", y = "Number of Employees", colour = "Industry") +
  theme_minimal(base_size = 20) +
  theme(plot.title = element_text(face = "bold", size = 26, hjust = 0.5),
        axis.title = element_text(face = "bold", size = 22),
        axis.text = element_text(size = 20),
        legend.title = element_text(size = 22, face = "bold"),
        legend.text = element_text(size = 20),
        legend.key.height = unit(1.2, "cm"),
        legend.key.width = unit(0.25, "cm"))
```

```
ggsave("employment_trends_all_industries.png", plot = employment_trends_all_industries,
width = 16, height = 12, dpi = 300)


#------------------------#
#     Top 5 Industries    #
#------------------------#
top_industries <- bres_long %>%
  group_by(Industry) %>%
  summarise(TotalEmployees = sum(Employees, na.rm = TRUE)) %>%
  arrange(desc(TotalEmployees)) %>%
  slice_head(n = 5)

bres_top <- bres_long %>% filter(Industry %in% top_industries$Industry)

ggplot(bres_top, aes(Year, Employees, colour = Industry)) +
  geom_line(size = 2) +
  geom_point(size = 4) +
  scale_y_continuous(labels = comma) +
  scale_x_continuous(breaks = 2015:2024) +
  labs(title = "Employment Trends for the Top 5 Industries (2015-2024)",
       x = "Year", y = "Number of Employees", colour = "Industry") +
  theme_minimal(base_size = 20) +
  theme(plot.title = element_text(face = "bold", size = 26, hjust = 0.5),
        axis.title = element_text(face = "bold", size = 22),
        axis.text = element_text(size = 20),
        legend.title = element_text(size = 22, face = "bold"),
        legend.text = element_text(size = 20),
        legend.key.height = unit(1.2, "cm"),
        legend.key.width = unit(0.25, "cm"))
```

```
ggsave("employment_trends_top5_industries.png", width = 16, height = 12, dpi = 300)


#=========================================================#
#                      RQ1                                #
#   Industries with the most consistent growth            #
#=========================================================#
industry_growth <- bres_long %>%
  group_by(Industry) %>%
  summarise(GrowthPercent = ((last(Employees) - first(Employees)) / first(Employees)) *
100) %>%
  arrange(desc(GrowthPercent))


ggplot(industry_growth, aes(reorder(Industry, GrowthPercent), GrowthPercent, fill =
GrowthPercent)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = round(GrowthPercent, 1)), hjust = -0.1, size = 5) +
  coord_flip() +
  scale_fill_gradient(low = "lightblue", high = "darkblue") +
  labs(title = "Percentage Employment Growth by Industry (2015-2024)",
       x = "Industry", y = "Growth (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 18),
        axis.text.y = element_text(size = 18),
        axis.title.x = element_text(size = 19, face = "bold"),
        axis.title.y = element_text(size = 21, face = "bold"),
        plot.title = element_text(face = "bold", hjust = 0.5, size = 24))


ggsave("percentage_growth_by_industry.png", width = 15, height = 12, dpi = 300)


#=========================================================#
```

```
#                      RQ2                                #
#   COVID-19 impact on employment (2019-2021)           #
#=========================================================#
covid_change <- bres_long %>%
  filter(Year %in% c(2019, 2021)) %>%
  group_by(Industry) %>%
  summarise(ChangeDuringCovid = ((last(Employees) - first(Employees)) / first(Employees)) *
100) %>%
  arrange(desc(ChangeDuringCovid))

ggplot(covid_change, aes(reorder(Industry, ChangeDuringCovid), ChangeDuringCovid, fill =
ChangeDuringCovid > 0)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = round(ChangeDuringCovid, 1)),
            hjust = ifelse(covid_change$ChangeDuringCovid > 0, -0.1, 1.1), size = 5) +
  scale_fill_manual(values = c("TRUE" = "forestgreen", "FALSE" = "firebrick")) +
  coord_flip() +
  labs(title = "Employment Change During COVID-19 (2019-2021)",
       x = "Industry", y = "Change (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 18),
        axis.text.y = element_text(size = 18),
        axis.title.x = element_text(size = 19, face = "bold"),
        axis.title.y = element_text(size = 21, face = "bold"),
        plot.title = element_text(face = "bold", hjust = 0.5, size = 24))

ggsave("covid_employment_change.png", width = 15, height = 12, dpi = 300)

#=========================================================#
#                      RQ3                                #
```

```
#   Predicting industry growth/decline by 2030           #
#========================================================#
fit_predict_industry <- function(df, year_pred = 2030) {
  model <- lm(Employees ~ Year, data = df)
  predicted <- predict(model, newdata = data.frame(Year = year_pred))
  tibble(
    Industry = unique(df$Industry),
    Employees_2024 = df %>% filter(Year == 2024) %>% pull(Employees),
    Predicted_2030 = round(predicted),
    Projected_Growth_Percent = ((predicted - Employees_2024) / Employees_2024) * 100
  )
}

predictions_2030 <- bres_long %>%
  group_by(Industry) %>%
  group_modify(~fit_predict_industry(.x)) %>%
  ungroup() %>%
  arrange(desc(Projected_Growth_Percent))

# Bar chart of projected growth
ggplot(predictions_2030, aes(reorder(Industry, Projected_Growth_Percent),
Projected_Growth_Percent, fill = Projected_Growth_Percent > 0)) +
  geom_col(show.legend = FALSE) +
  scale_fill_manual(values = c("TRUE" = "forestgreen", "FALSE" = "firebrick")) +
  coord_flip() +
  labs(title = "Predicted Employment Growth by Industry (2024-2030)",
       x = "Industry", y = "Projected Growth (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 18),
        axis.text.y = element_text(size = 18),
```

```
        axis.title.x = element_text(size = 19, face = "bold"),
        axis.title.y = element_text(size = 21, face = "bold"),
        plot.title = element_text(face = "bold", hjust = 0.5, size = 24))

ggsave("predicted_employment_change.png", width = 15, height = 12, dpi = 300)

# Historical and 2030 line plot
bres_long_2030 <- bres_long %>%
  bind_rows(predictions_2030 %>% transmute(Industry, Year = 2030, Employees =
Predicted_2030))

ggplot(bres_long_2030, aes(Year, Employees, colour = Industry)) +
  geom_line(size = 1.2) +
  geom_point(data = bres_long_2030 %>% filter(Year == 2030), size = 3, shape = 17) +
  scale_y_continuous(labels = comma) +
  labs(title = "Historical Employment Trends and 2030 Predictions",
       x = "Year", y = "Number of Employees") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 26, hjust = 0.5),
        axis.title = element_text(face = "bold", size = 22),
        axis.text = element_text(size = 20),
        legend.title = element_text(size = 22, face = "bold"),
        legend.text = element_text(size = 17),
        legend.key.height = unit(1.2, "cm"),
        legend.key.width = unit(1, "cm"),
        legend.position = "bottom") +
  guides(colour = guide_legend(ncol = 3))

ggsave("2030_predictions_line_graph.png", width = 16, height = 12, dpi = 300)
```

```
# Pie chart of predicted 2030 employment

predictions_2030_pie <- predictions_2030 %>%

  mutate(Share = Predicted_2030 / sum(Predicted_2030) * 100,

         Label = paste0(round(Share, 1), "%"))


ggplot(predictions_2030_pie, aes(x = 1, y = Share, fill = Industry)) +

  geom_col(color = "white") +

  coord_polar(theta = "y") +

  geom_text(aes(label = Label), position = position_stack(vjust = 0.5), size = 5, colour =
"black") +

  labs(title = "Predicted Employment Distribution by Industry (2030)", fill = "Industry") +

  theme_void() +

  theme(plot.title = element_text(face = "bold", size = 22, hjust = 0.5),

        legend.title = element_text(face = "bold", size = 20),

        legend.text = element_text(size = 16),

        legend.position = "right")


ggsave("predicted_2030_employment_piechart.png", width = 14, height = 12, dpi = 300)
```

8.2 APPENDIX B - GITHUB PROFILE AND PROJECT SCREENSHOTS

Figure B1

Screenshot of GitHub profile

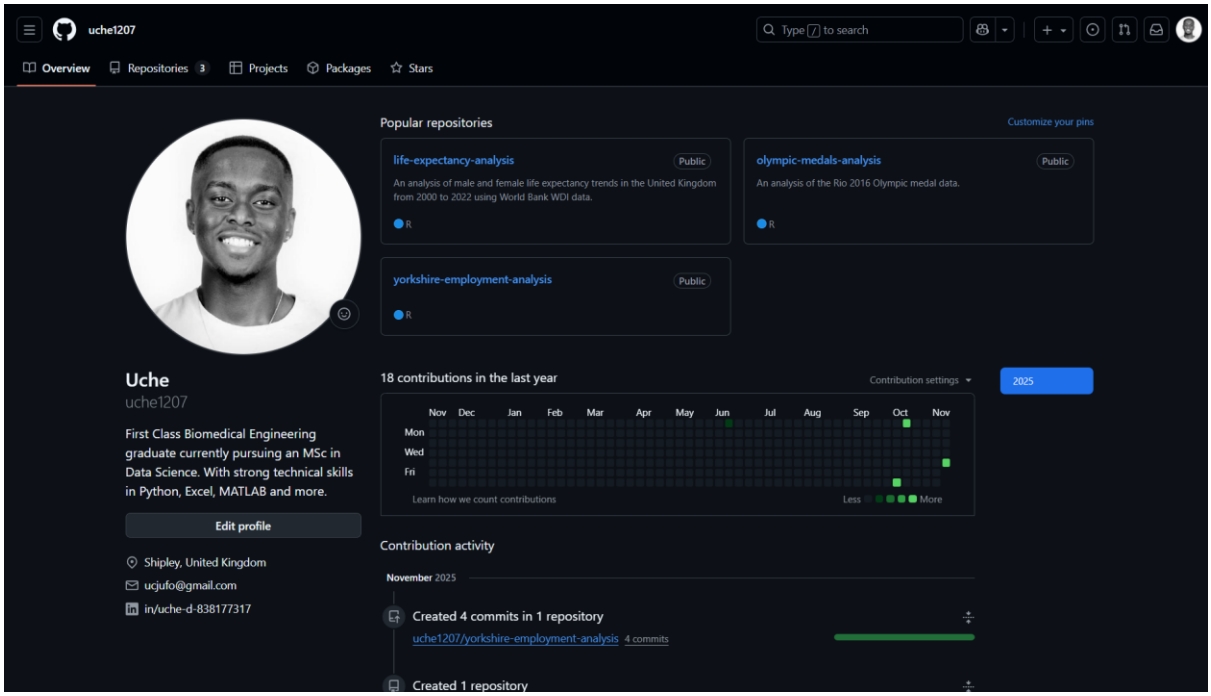


Figure B2

Screenshot of project repository

